PROGRAM:

AIES-Exp_3.py - C:/Users/ABBIRAMY/OneDrive/Documents/AIES_LAB/AIES-Exp_3.py (3.14.2)

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```
import heapq
grid = [
    [0, 0, 0, 0],
    [1, 1, 0, 1],
    [0, 0, 0, 0],
    [0, 1, 1, 0]
]   # 0 = free, 1 = blocked
def manhattan(a, b):
    return abs(a[0] - b[0]) + abs(a[1] - b[1])
def greedy(grid, start, goal):
    pq = [(manhattan(start, goal), start, [start])]
    visited = set()
    while pq:
        _, (r, c), path = heapq.heappop(pq)
        if (r, c) == goal:
            return path
        if (r, c) in visited:
            continue
        visited.add((r, c))
        for dr, dc in [(1, 0), (-1, 0), (0, 1), (0, -1)]:
            nr, nc = r + dr, c + dc
            if (0 <= nr < len(grid) and 0 <= nc < len(grid[0])
                    and grid[nr][nc] == 0 and (nr, nc) not in visited):
                heapq.heappush(pq, (manhattan((nr, nc), goal),
                                    (nr, nc), path + [(nr, nc)]))
    return None
print("Path:", greedy(grid, (0, 0), (3, 3)))
```

OUTPUT:

IDLE Shell 3.14.2

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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/ABBIRAMY/OneDrive/Documents/AIES_LAB/AIES-Exp_3.py =====
Path: [(0, 0), (0, 1), (0, 2), (1, 2), (2, 2), (2, 3), (3, 3)]
>>>
```